

Literature-Implied Answer: Strict H^∞ -Calculus Angle

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Source Question

The source paper is N. J. Kalton and L. Weis, *The H^∞ -calculus and sums of closed operators*, arXiv:math/0010155. In Section 4, after defining the sectorial angle $\omega(A)$ and the bounded H^∞ -calculus angle $\omega_H(A)$, the authors write:

Is $\omega_H(A) = \omega(A)$ whenever A is H^∞ -sectorial?

The sentence appears on page 16 of the source PDF.

Supporting Theorem

The supporting paper is N. J. Kalton, E. Lorist, and L. Weis, *Euclidean structures and operator theory in Banach spaces*, arXiv:1912.09347. Section 6.4 records the known counterexample lineage: Cowling–Doust–McIntosh–Yagi first constructed a bounded H^∞ -calculus example without dense range with $\omega(A) < \omega_{H^\infty}(A)$, and Kalton later constructed a sectorial example with the same strict inequality.

More decisively, Theorem 6.4.4 of arXiv:1912.09347 states that for every $p \in (1, \infty) \setminus \{2\}$ and every $\sigma \in (0, \pi)$, there is a closed subspace $Y \subset L^p([0, 1])$ and a sectorial operator A on Y such that A has a bounded H^∞ -calculus, has bounded imaginary powers, is almost γ -sectorial, and

$$\omega(A) = 0, \quad \omega_{H^\infty}(A) = \sigma.$$

Identification

The notation $\omega_H(A)$ in arXiv:math/0010155 is the same bounded H^∞ -calculus angle denoted $\omega_{H^\infty}(A)$ in the supporting paper. Since $\sigma > 0$ is arbitrary, Theorem 6.4.4 gives explicit H^∞ -sectorial operators for which the two angles are unequal. Therefore the source question has a negative literature answer.

Provenance and Scope

This is a literature-implied answer rather than an original proof. The supporting paper does not explicitly frame Theorem 6.4.4 as answering the Kalton–Weis sentence from arXiv:math/0010155; the implication follows by matching the angle notation and hypotheses. No broader claim is made here about all remaining functional-calculus or maximal-regularity questions.

References

- N. J. Kalton and L. Weis, *The H^∞ -calculus and sums of closed operators*, arXiv:math/0010155.
- N. J. Kalton, E. Lorist, and L. Weis, *Euclidean structures and operator theory in Banach spaces*, arXiv:1912.09347.
- N. J. Kalton, *A remark on sectorial operators with an H^∞ -calculus*, Contemp. Math. 321, 91–99, 2003.